

The module of derivations of graded Tjurina algebras

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The Tjurina Algebra

Let

$$S = \mathbb{C}[x_1, \dots, x_n].$$

For a polynomial $f \in S$, the *Jacobian ideal* of f is

$$J_f = \left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle.$$

Definition 1.1 (Tjurina algebra). The *Tjurina algebra* of f is

$$T(f) = \frac{\mathbb{C}[x_1, \dots, x_n]}{\langle f, J_f \rangle}.$$

The number

$$\tau(f) := \dim_{\mathbb{C}} T(f)$$

is called the *Tjurina number*.

Theorem 1.2 (Mather–Yau). *Let $f, g \in \mathbb{C}\{x_1, \dots, x_n\}$ define isolated hypersurface singularity germs at the origin. Then the germs*

$$(V(f), 0) \quad \text{and} \quad (V(g), 0)$$

are analytically isomorphic if and only if their local Tjurina algebras

$$\frac{\mathbb{C}\{x_1, \dots, x_n\}}{\langle f, J_f \rangle} \quad \text{and} \quad \frac{\mathbb{C}\{x_1, \dots, x_n\}}{\langle g, J_g \rangle}$$

are isomorphic as local $\mathbb{C}$ -algebras.

Remark 1.3. The Mather–Yau theorem says that the analytic isomorphism type of an isolated hypersurface singularity is completely determined by its Tjurina algebra. Thus the finite-dimensional algebra $T(f)$ packages the local analytic information of the singularity.

Yau’s Lie Algebra

Definition 1.4 (Yau algebra). Let $f \in \mathbb{C}[x_1, \dots, x_n]$ define an isolated hypersurface singularity. The *Yau algebra* associated to f is the Lie algebra

$$L(f) := \text{Der}_{\mathbb{C}}(T(f)).$$

Theorem 1.5 (Yau, 1981). *Let f define an isolated hypersurface singularity. Then*

$$L(f) = \text{Der}_{\mathbb{C}}(T(f))$$

is a finite-dimensional solvable Lie algebra.

Remark 1.6. The phrase in the handwritten notes that appears as “ $T(f)$ defines an isolated singularity which is solvable” is interpreted as the standard statement that the derivation Lie algebra of the Tjurina algebra of an isolated hypersurface singularity is solvable.

Program of Questions

The lecture notes outline the following sequence of questions.

- (1) Does the Lie algebra

$$L(f) = \text{Der}_{\mathbb{C}}(T(f))$$

distinguish isolated hypersurface singularities?

- (2) If $f \in \mathbb{C}[[x]]$ is weighted homogeneous, then $T(f)$ is graded and hence

$$L(f) = \bigoplus_{j \in \mathbb{Z}} L_j$$

is a graded Lie algebra.

- (3) Yau's conjectural direction concerns the negative-degree part of this graded Lie algebra. In the notation of the notes, this appears as

$$L_j \neq 0 \quad \text{for certain } j < 0.$$

- (4) If f is weighted homogeneous, then by Milnor–Orlik the dimension of the Tjurina algebra depends only on the weights:

$$\dim_{\mathbb{C}} T(f) \quad \text{depends only on the weights of } f.$$

- (5) Yau–Zuo type conjectures concern bounds for

$$\dim_{\mathbb{C}} \text{Der}_{\mathbb{C}}(L(f)).$$

- (6) Higher-order versions of these invariants and questions can be formulated.

Complex Hypersurfaces

Basic Definitions

Let

$$f \in \mathbb{C}[x_1, \dots, x_n].$$

The complex hypersurface defined by f is

$$V(f) = \{p \in \mathbb{C}^n \mid f(p) = 0\} \subsetneq \mathbb{C}^n.$$

Definition 1.7 (Singular locus). The singular locus of the hypersurface $V(f)$ is

$$\text{Sing}(f) = \left\{ p \in \mathbb{C}^n \mid f(p) = 0 \text{ and } \frac{\partial f}{\partial x_i}(p) = 0 \text{ for all } i = 1, \dots, n \right\}.$$

Equivalently,

$$\text{Sing}(f) = V(f, J_f).$$

Definition 1.8 (Isolated hypersurface singularity). The hypersurface $V(f)$ has an *isolated hypersurface singularity* at the origin if

$$\text{Sing}(f) = \{\underline{0}\}$$

in a sufficiently small analytic neighborhood of the origin.

Examples

Example 1.9 (The cusp). The plane curve

$$V(x^3 - y^2) \subseteq \mathbb{C}^2$$

has a cusp singularity at the origin.

Example 1.10 (The quadratic cone). The hypersurface

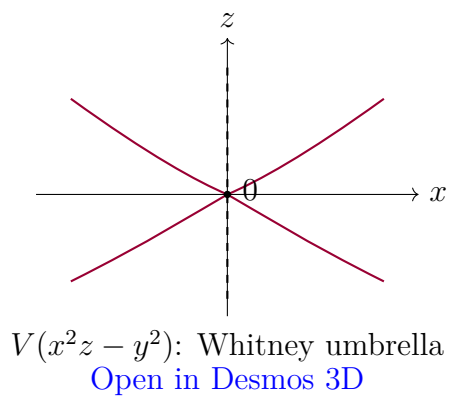
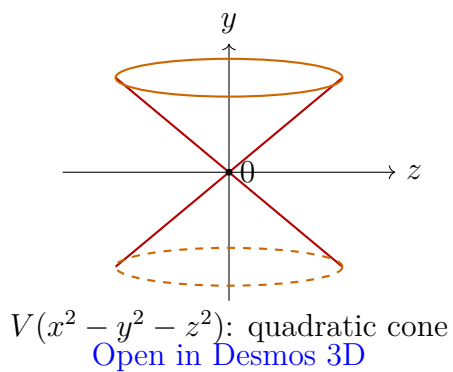
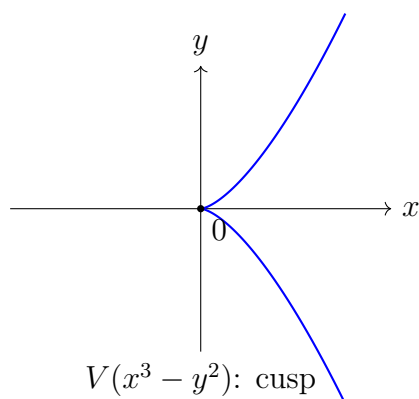
$$V(x^2 - y^2 - z^2) \subseteq \mathbb{C}^3$$

is a cone with an isolated singularity at the origin.

Example 1.11 (The Whitney umbrella). The surface

$$V(x^2z - y^2) \subseteq \mathbb{C}^3$$

is the Whitney umbrella.



Milnor and Tjurina Algebras

Definition 1.12 (Milnor algebra). The *Milnor algebra* of f is

$$M(f) = \frac{\mathbb{C}[x_1, \dots, x_n]}{J_f} = \frac{\mathbb{C}[x_1, \dots, x_n]}{\left\langle \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle}.$$

Its complex vector space dimension

$$\mu(f) := \dim_{\mathbb{C}} M(f)$$

is called the *Milnor number*.

Definition 1.13 (Tjurina algebra). The *Tjurina algebra* of f is

$$T(f) = \frac{\mathbb{C}[x_1, \dots, x_n]}{\langle f, J_f \rangle} = \frac{\mathbb{C}[x_1, \dots, x_n]}{\left\langle f, \frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right\rangle}$$

Its complex vector space dimension

$$\tau(f) := \dim_{\mathbb{C}} T(f)$$

is called the *Tjurina number*.

Remark 1.14. The Milnor number $\mu(f)$ measures the topology of the Milnor fiber. For an isolated hypersurface singularity, the Milnor fiber has the homotopy type of a bouquet of $\mu(f)$ spheres. The Tjurina number $\tau(f)$ measures the dimension of the base of the semi-universal deformation. If f is weighted homogeneous of weighted degree d with weights $w_1, \dots, w_n$, and if f has an isolated singularity at the origin, then

$$\mu(f) = \tau(f).$$

Moreover, the Milnor number is determined by the weight system and the weighted degree. Equivalently, after normalizing the weights by d , it depends only on the normalized weights.

Derivations

Derivations of Algebras

Definition 1.15 (Derivation). Let R be a k -algebra. A k -derivation of R is a k -linear map

$$D : R \rightarrow R$$

satisfying the Leibniz rule

$$D(ab) = aD(b) + bD(a)$$

for all $a, b \in R$. The set of all k -derivations of R is denoted by

$$\text{Der}_k(R).$$

Example 1.16. Let

$$R = \mathbb{C}[x_1, \dots, x_n].$$

Then each partial derivative

$$\partial_{x_i} : R \rightarrow R$$

is a $\mathbb{C}$ -derivation. Moreover, every $\mathbb{C}$ -derivation $D : R \rightarrow R$ can be written uniquely as

$$D = \sum_{i=1}^n D(x_i) \partial_{x_i}.$$

Therefore

$$\text{Der}_{\mathbb{C}}(\mathbb{C}[x_1, \dots, x_n]) \cong \bigoplus_{i=1}^n \mathbb{C}[x_1, \dots, x_n] \partial_{x_i}$$

is a free R -module of rank n .

Derivations on Quotients

Question 1.17. How can one describe

$$\text{Der}_{\mathbb{C}} \left(\frac{\mathbb{C}[x_1, \dots, x_n]}{I} \right)?$$

A naive attempt is to descend the usual partial derivatives to the quotient. This does not always work.

Example 1.18. Let

$$A = \frac{\mathbb{C}[x]}{\langle x^2 \rangle}.$$

The usual derivative

$$\partial_x : \mathbb{C}[x] \rightarrow \mathbb{C}[x]$$

does not induce a well-defined derivation on A . Indeed,

$$[x^2] = [0] \in A,$$

but

$$\partial_x(x^2) = 2x,$$

and

$$[2x] \neq [0] \in A.$$

Equivalently,

$$\partial_x(\langle x^2 \rangle) \not\subseteq \langle x^2 \rangle.$$

Definition 1.19 (Derivations preserving an ideal). Let $I \subseteq \mathbb{C}[x_1, \dots, x_n]$. Define

$$\text{Der}_I(\mathbb{C}[x_1, \dots, x_n]) = \{D \in \text{Der}_{\mathbb{C}}(\mathbb{C}[x_1, \dots, x_n]) \mid D(I) \subseteq I\}.$$

These are the derivations preserving the ideal I .

Proposition 1.20. *Let*

$$S = \mathbb{C}[x_1, \dots, x_n]$$

and let $I \subseteq S$ be an ideal. Then there is a canonical isomorphism

$$\text{Der}_{\mathbb{C}} \left(\frac{S}{I} \right) \cong \frac{\text{Der}_I(S)}{I \text{Der}_{\mathbb{C}}(S)}.$$

Proof. Every derivation $D \in \text{Der}_I(S)$ induces a derivation

$$\overline{D} : S/I \rightarrow S/I$$

by

$$\overline{D}([f]) = [D(f)].$$

This is well-defined precisely because $D(I) \subseteq I$.

Thus we obtain a natural map

$$\text{Der}_I(S) \longrightarrow \text{Der}_{\mathbb{C}}(S/I).$$

Its kernel consists of those derivations D such that

$$D(S) \subseteq I.$$

Since

$$\text{Der}_{\mathbb{C}}(S) = \bigoplus_{i=1}^n S \partial_{x_i},$$

this kernel is exactly

$$I \text{Der}_{\mathbb{C}}(S).$$

Therefore the induced map gives the desired isomorphism

$$\frac{\text{Der}_I(S)}{I \text{Der}_{\mathbb{C}}(S)} \cong \text{Der}_{\mathbb{C}}(S/I).$$

□

$$\begin{array}{ccc} S & \xrightarrow{D} & S \\ \downarrow & & \downarrow \\ S/I & \xrightarrow{\overline{D}} & S/I \end{array}$$

Lie Algebras

Basic Definitions

Definition 1.21 (Lie algebra). A Lie algebra over $\mathbb{C}$ is a $\mathbb{C}$ -vector space L together with a bilinear map

$$[-, -] : L \times L \rightarrow L, \quad (X, Y) \mapsto [X, Y],$$

called the *Lie bracket*, satisfying:

i) $[X, X] = 0$ for all $X \in L$.

ii) The Jacobi identity:

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$$

for all $X, Y, Z \in L$.

Remark 1.22. Since the base field is $\mathbb{C}$, the condition $[X, X] = 0$ is equivalent to skew-symmetry:

$$[X, Y] = -[Y, X].$$

Example 1.23 (Derivations form a Lie algebra). Let A be a $\mathbb{C}$ -algebra. Then

$$\mathrm{Der}_{\mathbb{C}}(A)$$

is a Lie algebra with bracket

$$[D, E] = D \circ E - E \circ D.$$

Definition 1.24 (Homomorphism of Lie algebras). Let L_1 and L_2 be Lie algebras. A map

$$\Phi : L_1 \rightarrow L_2$$

is a homomorphism of Lie algebras if:

i) Φ is $\mathbb{C}$ -linear.

ii) For all $X, Y \in L_1$,

$$\Phi([X, Y]_{L_1}) = [\Phi(X), \Phi(Y)]_{L_2}.$$

If Φ is bijective, then Φ is an isomorphism of Lie algebras.

Derivation Lie Algebras of Tjurina Algebras

Let $f \in \mathbb{C}[x_1, \dots, x_n]$ define an isolated hypersurface singularity. The central algebraic object in these notes is

$$L(f) = \mathrm{Der}_{\mathbb{C}}(T(f)).$$

Since

$$T(f) = \frac{S}{\langle f, J_f \rangle},$$

the quotient description of derivations gives

$$L(f) \cong \frac{\mathrm{Der}_{\langle f, J_f \rangle}(S)}{\langle f, J_f \rangle \mathrm{Der}_{\mathbb{C}}(S)}.$$

Remark 1.25. This formula allows one to compute $L(f)$ explicitly. A derivation

$$D = \sum_{i=1}^n a_i \partial_{x_i}$$

descends to $T(f)$ precisely when it preserves the Tjurina ideal:

$$D(\langle f, J_f \rangle) \subseteq \langle f, J_f \rangle.$$

Weighted Homogeneous Case

Definition 1.26 (Weighted homogeneous polynomial). Let $w_1, \dots, w_n$ be positive rational weights. A polynomial

$$f \in \mathbb{C}[x_1, \dots, x_n]$$

is weighted homogeneous of weighted degree d if each monomial

$$x_1^{a_1} \cdots x_n^{a_n}$$

appearing in f satisfies

$$a_1 w_1 + \cdots + a_n w_n = d.$$

Remark 1.27. If f is weighted homogeneous, then the Euler derivation

$$E = \sum_{i=1}^n w_i x_i \partial_{x_i}$$

satisfies the Euler identity

$$E(f) = df.$$

Consequently,

$$f \in J_f$$

whenever the weighted degree d is nonzero in the base field. Over $\mathbb{C}$, this always holds. Thus, in the weighted homogeneous isolated hypersurface case,

$$T(f) = M(f).$$

After clearing denominators, we may assume the weights $w_1, \dots, w_n$ are positive integers. Then S , $T(f)$, and

$$L(f) = \text{Der}_{\mathbb{C}}(T(f))$$

inherit $\mathbb{Z}$ -gradings:

$$L(f) = \bigoplus_{j \in \mathbb{Z}} L_j.$$

Absolutely Isolated Surface Singularities

Double Points

Definition 1.28 (Double point). Let

$$f = f_2 + f_3 + f_4 + \dots$$

be the decomposition of a convergent power series into homogeneous parts, and assume $f(0) = 0$ and $df(0) = 0$. If

$$f_2 \neq 0,$$

then the hypersurface singularity $V(f)$ at the origin has multiplicity 2 and is called a double point.

Definition 1.29 (Absolutely isolated singularity). Let

$$X = V(f) \subseteq \mathbb{C}^n$$

and suppose $0 \in X$ is an isolated singularity. The singularity is called *absolutely isolated* if a resolution of singularities of X can be obtained by blowing up only points.

Example 1.30. The surface

$$V(x^2 - y^2 - z^2)$$

is an absolutely isolated singularity.

Example 1.31 (The Whitney umbrella). The surface

$$V(x^2 z - y^2) \subseteq \mathbb{C}^3$$

is the Whitney umbrella. Its singular locus is the line

$$\{x = 0, y = 0\}.$$

Thus it is not an isolated hypersurface singularity.

ADE Classification

Theorem 1.32 (ADE classification of rational double points). *Let*

$$X = V(f) \subseteq \mathbb{C}^3$$

be a normal surface singularity at the origin. Then X is a rational double point, equivalently a Du Val singularity, if and only if it is analytically isomorphic to exactly one of the following hypersurface singularities:

$$A_k : \quad x^{k+1} + y^2 + z^2 = 0, \quad k \geq 1,$$

$$D_k : \quad x^{k-1} + xy^2 + z^2 = 0, \quad k \geq 4,$$

$$E_6 : \quad x^3 + y^4 + z^2 = 0,$$

$$E_7 : \quad x^3 + xy^3 + z^2 = 0,$$

$$E_8 : \quad x^3 + y^5 + z^2 = 0.$$

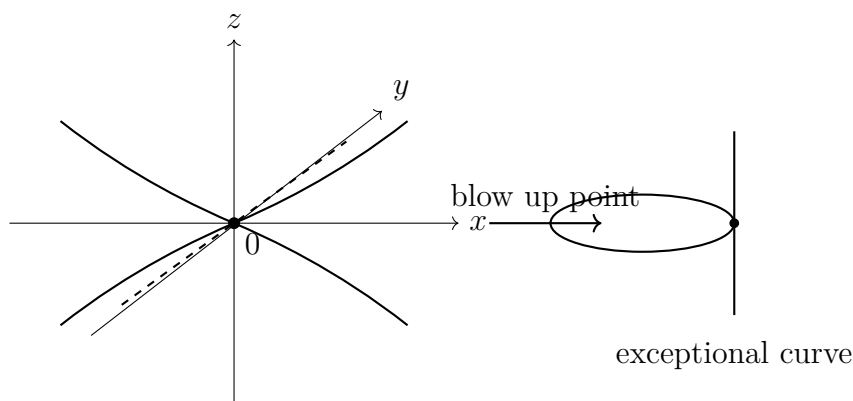
Remark 1.33. The ADE singularities are also called simple surface singularities or rational double points. Their classification is governed by the simply-laced Dynkin diagrams

$$A_k, \quad D_k, \quad E_6, \quad E_7, \quad E_8.$$

These diagrams also appear as the dual intersection graphs of exceptional divisors in the minimal resolution.

Schematic Picture

The following schematic diagram represents the idea that a double point surface singularity has a distinguished singular point at the origin, and resolution may proceed by blowing up points in the absolutely isolated case.



Summary of the Lecture

The notes establish the following chain of ideas.

- (1) A hypersurface singularity

$$V(f) \subseteq \mathbb{C}^n$$

has an associated Jacobian ideal J_f .

(2) The Tjurina algebra

$$T(f) = \frac{\mathbb{C}[x_1, \dots, x_n]}{\langle f, J_f \rangle}$$

is finite-dimensional when f defines an isolated hypersurface singularity.

(3) By the Mather–Yau theorem, the Tjurina algebra determines the analytic isomorphism class of the isolated hypersurface singularity.

(4) The derivation Lie algebra

$$L(f) = \text{Der}_{\mathbb{C}}(T(f))$$

is a secondary invariant attached to the singularity.

(5) For weighted homogeneous singularities, $L(f)$ inherits a grading

$$L(f) = \bigoplus_j L_j.$$

(6) In surface singularity theory, absolutely isolated double points are classified by the ADE list:

$$A_k, \quad D_k, \quad E_6, \quad E_7, \quad E_8.$$